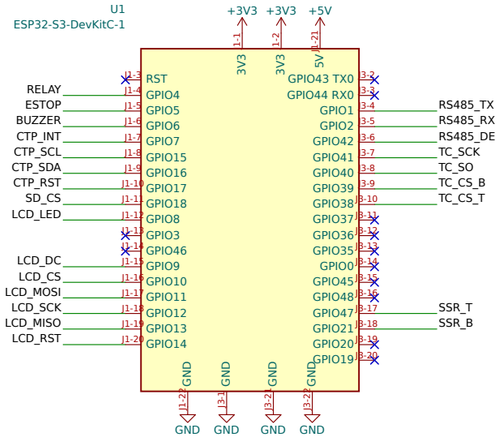
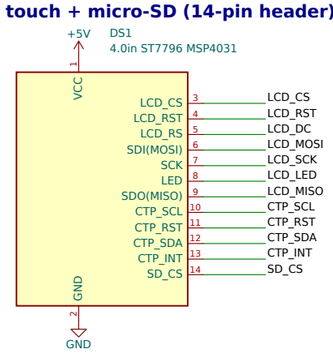
ESP32-S3-DevKitC-1 (v1.0, N16R8)



J1 = left header, J3 = right header (top view, USB ports at the bottom).
5 V from USB (J1-21), 3V3 from the DevKit's regulator (J1-1 / J1-2).
Not connected: GPIO35-37 (octal PSRAM), 19/20 and 43/44 (both USB ports), strapping pins 0, 3, 45, 46; GPIO48 = on-board status LED.

PWR_FLAG — one ground for the DevKit and every module

Display + touch + micro-SD (14-pin header)



VCC 5 V (backlight about 100 mA), logic 3.3 V (level shifters on the module).
Header pin numbers: check them against the print on the module.
The micro-SD slot (run logs) shares SCK / MOSI / MISO; SD_CS selects it.

Notes

TempCtl Duo, env s3; pins as in src/board_s3.h (firmware 1.0.38).
The modules are connected with cables: this is the connection schematic of the whole device, not a PCB layout.

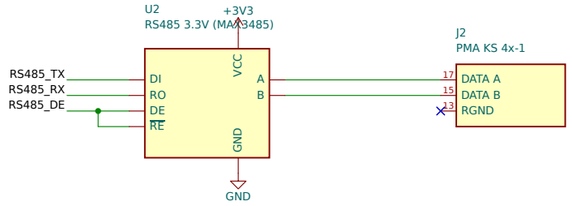
Net labels with the same name are connected.
+5V_SW = the relay-switched +5 V of both SSR inputs.

Display SPI 40 MHz on GPIO 9-14: keep these wires short.
SD card 10 MHz on the same bus, touch I2C 400 kHz, thermocouples bit-banged on their own bus (GPIO 38-41).

RS485: only 3.3 V modules (a 5 V MAX485 drives 5 V into RX).

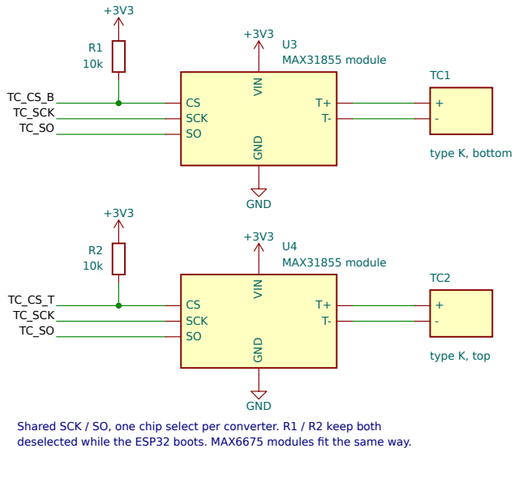
A software controller is never the only protection: fit an independent over-temperature limiter and a contactor.

RS485 (Modbus RTU, beta)



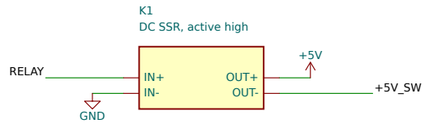
One or two controllers on the same A / B pair (own addresses).
No answer: swap A / B. No 120 ohm terminators below 8 devices.
RGND not connected. Auto-direction module: DE / RE open.

Thermocouples (local channels)



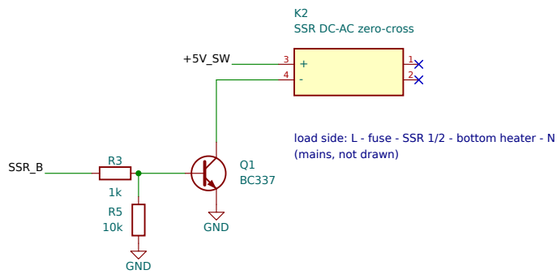
Shared SCK / SO, one chip select per converter. R1 / R2 keep both deselected while the ESP32 boots. MAX6675 modules fit the same way.

SSR supply relay (switched +5 V of both SSR inputs)



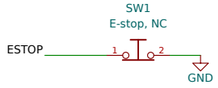
Active-high DC solid-state relay (input 3-32 V DC), driven directly by GPIO4: input current at most 20 mA at 3.3 V, output at least 0.5 A DC. GPIO4 is low (off) from the first line of setup(). On while a local channel heats, off at once on a fault or E-stop. It covers a stuck SSR input or driver; a shorted SSR output needs a contactor in the heater supply.

SSR drivers and SSRs (local channels)



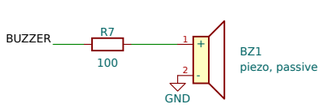
Q1 / Q2: BC337 or 2N2222 (TO-92 C-B-E). The 10k base pull-down keeps the SSR off during reset and flashing.
SSR + inputs: the relay-switched +5 V (+5V_SW).

E-stop input



Normally closed contact to GND, internal pull-up. Open contact or broken wire: SSRs + relay off, run stopped (setting safety.estop, off by default).

Alarm buzzer



2.7 kHz square wave (setting safety.sound). Passive piezo: directly via R7.

Magnetic buzzer / small speaker: NPN driver like Q1 (1k base) + 1N4148 across it.

Modules connected with cables; pins: src/board_s3.h

Title: **TempCtl Duo - ESP32-S3 wiring**

env s3, firmware 1.0.38

Size: A3 | Date: 2026-10-13 | Rev: 1.0.38 | Sheet: 1/1

File: TempCtl-Duo-S3.kicad_sch